

Sensor Chosen: MQ – 135 Gas Sensor.

Justifications:-

1. *Are we looking for atmospheric signatures?*

Yes. We are interested in identifying gases like ammonia, methane, and CO₂ that might be indicative of microbial life. The MQ – 135 sensor is capable of detecting most of these gases.

2. *Could this sensor tell us more about the habitability conditions of the site?*

Yes. Atmospheric composition is a key factor in determining habitability. For example, high levels of CO₂ without corresponding O₂ or methane might indicate a harsh environment. So, this sensor helps in mapping gas concentrations, thereby contributing to habitability assessments.

3. *If not for habitability factors, can this sensor be used elsewhere in our rover?*

Yes.

- a. Monitoring rover's internal air quality (for manned rovers).
- b. Checking for leaks or contamination in compartments.
- c. Correlating gas data with soil analysis result (e.g., the Bradford test).

References:-

<https://www.olimex.com/Products/Components/Sensors/Gas/SNS-MQ135/resources/SNS-MQ135.pdf>

<https://www.science.org/doi/10.1126/sciadv.abn7412>